Xenon:

Xenon is used in fluorescent bulbs, flash bulbs and lasers.

Xenon emits an intense light in discharge tubes instantly. Due to this it is used in high speed electronic flash bulbs used by photographers

Radon:

Radon is radioactive and used as a source of gamma rays

Radon gas is sealed as small capsules and implanted in the body to destroy malignant i.e. cancer growth

Summary

- **Occurrence:** About 78 % of earth atmosphere contains dinitorgen (N_2) gas. It is also present in earth crust as sodium nitrate (Chile saltpetre) and potassium nitrates (Indian saltpetre).
- **Nitrogen**, the principle gas of atmosphere (78 % by volume) is separated industrially from liquid air by fractional distillation
- Ammonia is formed by the hydrolysis of urea.
- Nitric acid is prepared by heating equal amounts of potassium or sodium nitrate with concentrated sulphuric acid.
- In most of the reactions, nitric acid acts as an oxidising agent. Hence the oxidation state changes from +5 to a lower one. It doesn't yield hydrogen in its reaction with metals.
- The reactions of metals with nitric acid are explained in 3 steps as follows:
  - **Primary reaction:** Metal nitrate is formed with the release of nascent hydrogen
  - **Secondary reaction:** Nascent hydrogen produces the reduction products of nitric acid.
  - **Tertiary reaction:** The secondary products either decompose or react to give final products
- Phosphorus has several allotropic modification of which the three forms namely white, red and black phosphorus are most common.
- Yellow phosphorus is poisonous in nature and has a characteristic garlic smell. It glows in the dark due to oxidation which is called phosphorescence.
- Yellow phosphorus readily catches fire in air giving dense white fumes of phosphorus pentoxide.
- Phosphine is prepared by action of sodium hydroxide with white phosphorous in an inert atmosphere of carbon dioxide or hydrogen.

XII U3-P-block.indd 94 16-12-2022 18:37:20